Theory Assignment

Roll No: 24M0326, 24M0319

Let $f(x)$ and $g(x)$ are two convex functions

Assume $h(x) = f(x) - g(x)$ is also convex, then

$$h(\lambda x + (1-\lambda) y) \leq \lambda h(x) + (1-\lambda) h(y) \quad , \lambda \in [0, 1]$$

$$\Rightarrow f(\lambda x + (1-\lambda) y) - g(\lambda x + (1-\lambda) y)$$

$$\leq \lambda (f(x) - g(x)) + (1-\lambda) (f(y) - g(y))$$

$$\Rightarrow f(\lambda x + (1-\lambda) y) - g(\lambda x + (1-\lambda) y)$$

$$\leq \lambda f(x) + (1-\lambda) f(y)$$

$$- [\lambda g(x) + (1-\lambda) g(y)]$$

$$\Rightarrow f(\lambda x + (1-\lambda) y) - \lambda f(x) - (1-\lambda) f(y)$$

$$- [g(\lambda x + (1-\lambda) y) - \lambda g(x) - (1-\lambda) g(y)] \leq 0$$

$$\Rightarrow \boxed{D_f - D_g \leq 0} \rightarrow (1)$$

where $D_f = f(\lambda x + (1-\lambda) y) - \lambda f(x) - (1-\lambda) f(y)$

Since both f and g are convex

$D_f \leq 0$ and $D_g \leq 0$

i.e. D_f and D_g are non positive

But difference of two non-positive ~~is no~~ quantity is not necessarily a non-positive quantity.

So, eqn (1) is ~~ϕ~~ not ~~true~~ necessarily true.

which in turn implies $h(x) = f(x) - g(x)$ is not necessarily convex.